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Lecture Notes «Matemātiskā analīze I»

CALCULUS I

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LECTURE 4.

CONTINUOUS FUNCTIONS

Basic definitions
Continuity of the elementary functions
Points of discontinuity
Properties of the continuous functions
4. Homework

Basic definitions

In math an science, we use word *continuous* to describe process that goes on without abrupt changes.
Our experiece leads us to assume that this is an essential feature of many natural processes.

DEFINITION 4.1. (1st definition of the continuity, continuity at a point)
Let *f(x)* be defined on an open interval containing x_0. We say that *f(x)* is continuous at x_0 if

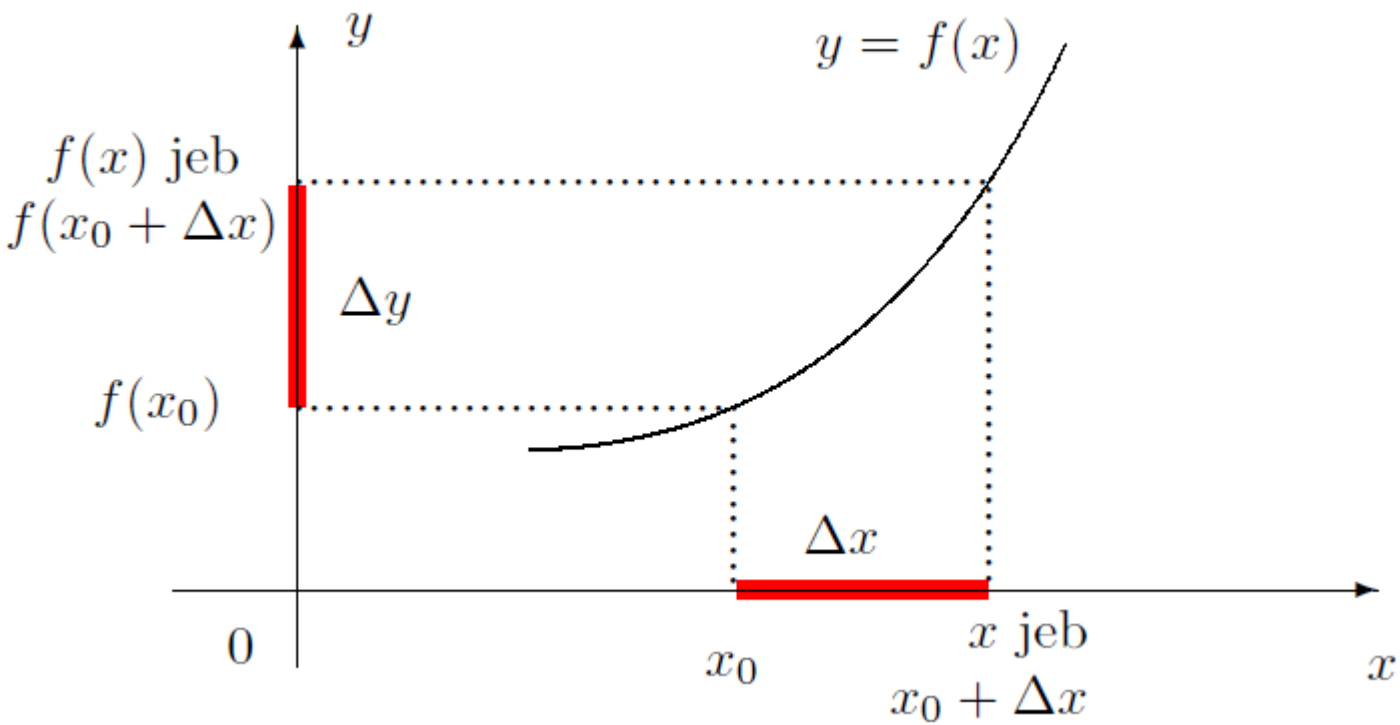
$$\lim_{x \to x_0} f(x) = f(x_0)$$

We mean by this definition to require three things:

1. $\lim_{x \to x_0} f(x)$ exists,
2. $f(x_0)$ exists (x_0 is in the domain of *f*), and
3. $\lim_{x \to x_0} f(x) = f(x_0)$

If any one of these three fails, then *f(x)* is discontinuous at x_0.

Basic definitions



$\Delta x = x - x_0$ – change in x
$\Delta y = f(x) - (x_0)$ – change in y (denotation: also $\Delta f(x_0)$)

Note that $\Delta y = f(x) - f(x_0) = f(x_0 + \Delta x) - f(x_0)$.
Thus, Δy can be seen also as function of argument Δx.

Basic definitions

DEFINITION 4.2. (2nd definition of the continuity)
We say that *f(x)* is continuous at x_0 if

$$\lim_{\Delta x \to 0} \Delta y = 0$$

DEFINITION 4.3. We say that function *u(x)* is infinitely-small function as *x* approaches x_0 if

$$\lim_{x \to x_0} u(x) = 0$$

The second definition of the continuity shows that Δy(or $\Delta f(x_0)$) is infinitely-small function as Δx approaches 0. In such case, we can say that for infinitely-small change in argument Δx at x_0 corresponds infinitely-small change in function $\Delta f(x_0)$.

Basic definitions

DEFINITION 4.4. We say that function *f(x)* is left continuous at x_0 if $\lim_{x \to x_0^-} f(x) = f(x_0)$. We say that function *f(x)* is right continuous at x_0 if $\lim_{x \to x_0^+} f(x) = f(x_0)$.

DEFINITION 4.5. (3rd definition of the continuity)
We say that function *f(x)* is continuous at x_0 if function is left continuous at x_0 and right continous at x_0.

4.1., 4.2. and 4.5. definitions are equivalent.

We call function *f(x)* continuous in a set X (on an open interval), if *f(x)* is continuous in each point of the set X (of that interval). It is continuous on the closed interval [a,b] if it is continuous on (a,b), right continuous at a, and left continuous at b.

Continuity of the elementary functions

From main limit theorem we get a such theorem.

Theorem 4.6. (Continuity under function operations)
If functions *f(x)* and *g(x)* are continuous at x_0, then so are *kf(x)*, *f(x)*+*g(x)*, *f(x)*-*g(x)*, *f(x)***g(x)*, *f(x)*/*g(x)* (provided that $g(x_0) \neq 0$), and $\sqrt[n]{f(x)}$(provided that $(f(x_0) > 0$ if *n* is even).

Examples

1.

$f(x) = c, c - const$, for each x_0 the $\lim\limits_{x \to x_0} f(x) = c$. That means that the constant function is continous at every real number.

2.

$f(x) = x$, for each $x_0 \in \mathbb{R}$ the $\lim\limits_{x \to x_0} f(x) = x_0$. That means that the function $f(x) = x$ is continous at every real number.

Continuity of the elementary functions

Since any rational function is obtained by adding, multiplying and dividing continuous functions f(x)=c and g(x)=x, then the rational function is continuous at any point in their domain.

The sine, cosine, and 2^x functions are continuous at every real number x_0.
<u>In general, a larger result is valid.</u>

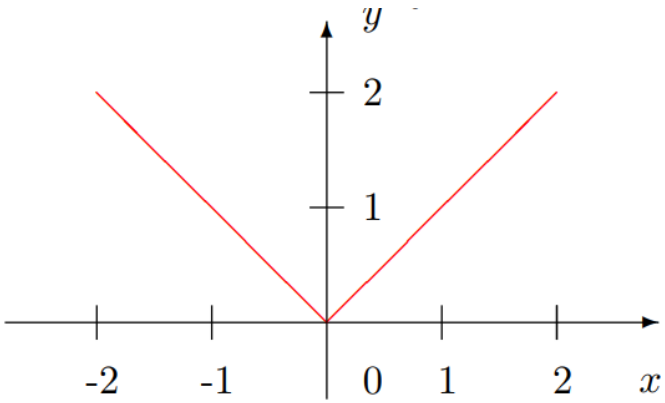
Theorem 4.7.
Any elementary function *f(x)* is continuous in its domain *Dom(f)*.

Continuity of the elementary functions

Example

Function $y = |x|$ is continuous at every real number, even if it is not elementary function.

$$|x| = \begin{cases} x, x \geq 0 \\ -x, x < 0 \end{cases}$$



For $x < 0, y = -x,$ a polynomial; for $x > 0, y = x,$ another polynomial.
Both are continuous in their domains.

$$\lim_{x \to 0^-} (-x) = 0\,,\ \lim_{x \to 0^+} x = 0\,, that\ is \lim_{x \to 0} |x| = 0$$

Therefore, |x| is continuous at 0.
We can see that |x| is continuous at every real number.

Continuity of the elementary functions

Theorem 4.8. (Composite limit theorem)
If *g(y)* is continuous at a point y_0 and $\lim_{x \to x_0} f(x) = y_0$, then

$$\lim_{x \to x_0} (g \circ f)(x) = g(\lim_{x \to x_0} f(x))$$

Result
If *g(y)* is continuous at a point y_0, but *f(x)* is continuous at a point x_0 and $f(x_0) = y_0$, then composition of the functions $(g \circ f)(x)$ is continuous at x_0.

Thus, we can also conclude: if *g(y)* is continuous at every real number, but *f(x)* is continuous in a set X, then $(g \circ f)(x)$ is continuous in the set X.

Example
$h(x) = |x^2 - 3x + 8|$ is continuous at every $x \in \mathbb{R}$, because $f(x) = |x| and\ g(x) = x^2 - 3x + 8$ are continuous functions. Thus their composition $h(x) = f\big(g(x)\big) = (f \circ g)(x)$ is continuous.

Points of discontinuity

We will consider classification of such points x_0 for which the deleted neighbourhood $\dot{\mathfrak{A}}(x_0, \delta_0)$ includes infinite amount of points from function *f(x)* domain *Dom(f)*. ($\dot{\mathfrak{A}}(x_0, \delta_0) \subseteq Dom(f)$)

DEFINITION 4.9. The point x_0 is a point of discontinuity if $x_0 \notin Dom(f)$ or the function is not continuous at this point.

If at the point of discontinuity x_0 exist one-sided limits, then such a point is called discontinuity of the first kind (or jump discontinuity, step discontinuity). At this situation the difference $\lim\limits_{x \to x_0^+} f(x) - \lim\limits_{x \to x_0^-} f(x)$ is called a jump at x_0.

If the jump is 0, then the point of discontinuity x_0 is called removable point of discontinuity. Otherwise the point of discontinuity is called nonremovable.

If a point of discontinuity x_0 is not a discontinuity of the first kind, then it is called discontinuity of the second kind (or essential discontinuity, infinite discontinuity).

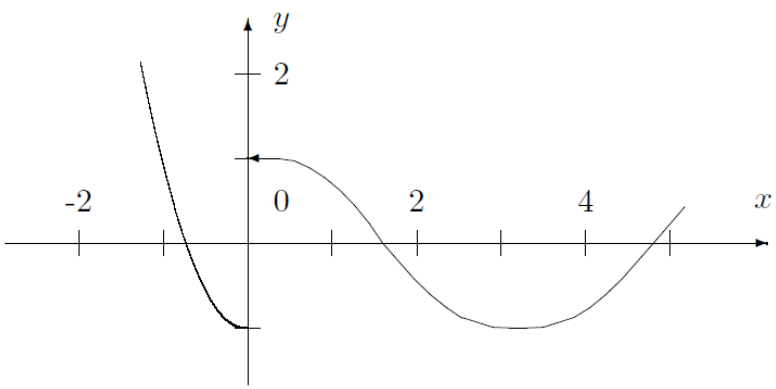
Points of discontinuity

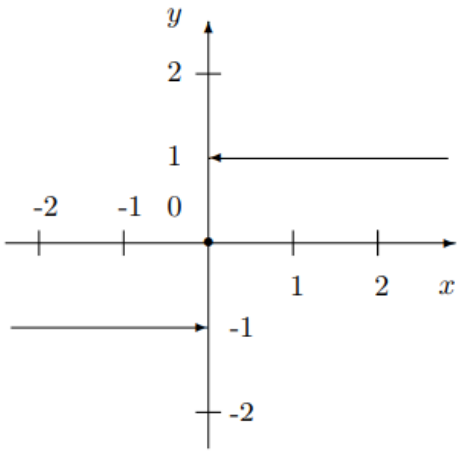
Examples

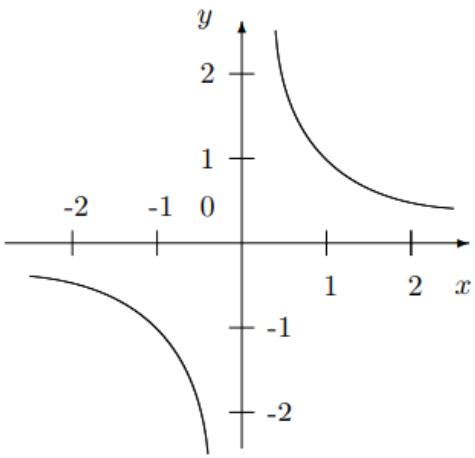
Please, indicate the points of discontinuity!

$y = \begin{cases} 2x^2 - 1, x \leq 0 \\ cosx, x > 0 \end{cases}$, $y = sgn(x)$(signum from x), and $y = \frac{1}{x}$

1. discontinuity of the first kind with a jump 2, at 0 the function is continuous from left-hand side
2. At 0 the function does not have one sided limits
3. discontinuity of the second kind



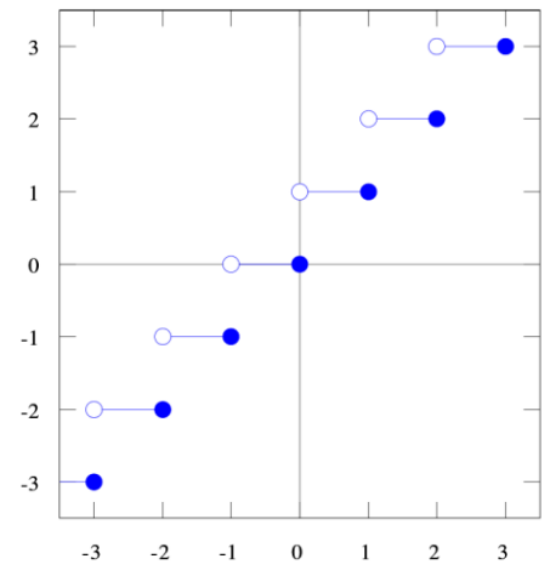




Points of discontinuity

Examples

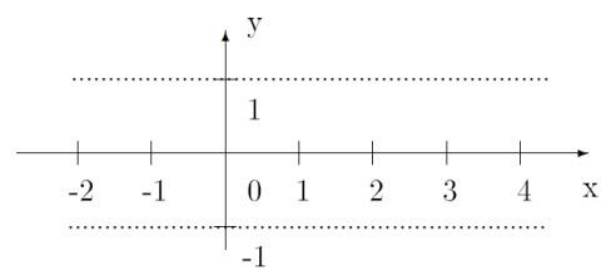
Integer function $\lfloor x \rfloor$ is right continuous at every real number, even if at $x_0 \in \mathbb{Z}$ it has discontinuity of the first kind (tha jump is 1).



For function
$y = \begin{cases} 1, x \neq 0 \\ -1, x = 0 \end{cases}$ the point 0 is a removable point of discontinuity, as well as for function
$y = \frac{x^2-8}{x-2}$ the point 2 is a removable point of discontinuity, even if the function is not Defined at this point.

In turn, for Dirichlet Function all points are discontinuity of the second kind, even if domain of the function include every real number.

$$D(x) = \begin{cases} -1 & , \text{ ja } x \in \mathbb{Q}; \\ 1 & , \text{ ja } x \notin \mathbb{Q}. \end{cases}$$



Properties of the continuous functions

DEFINITION 4.10. Lets say that $x_0 \in Dom(f)$ and $S \subseteq Dom(f)$.
f(c) is called the greatest or maximum value (element) of function *f* in a set S if $f(c) \geq f(x)$ for each $x \in S$.
f(c) is called the least or minimum value (element) of function *f* in a set S if $f(c) \leq f(x)$ for each $x \in S$.

The maximal and minimal values of the function *f* in a set S are called extrema.

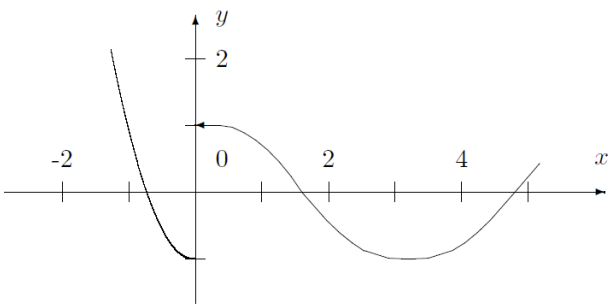
Often the existence of maximum and minimum for function is dependant from the set S.
For example, for function $f(x) = \frac{1}{x}$ in a set $S = (0; +\infty)$ do not have extremum. However, if $S = [a; b], a > 0$, then f(a) is maximum, but f(b) is minimum of the function *f*. In turn, if $S = (a; b)$, then again the function *f* does not have extremum.

Properties of the continuous functions

The existence of extremum is dependant from the type of the function.
If we consider discontinuous function

$$y = \begin{cases} 2x^2 - 1., x \leq 0 \\ cosx, x > 0 \end{cases}$$

then in a set $S = \left[-\frac{1}{2}; 4\right]$ the function does not have maximum, but it has minumum $y(0) = y(\pi) = -1$.



DEFINITION 4.11. Function f(x) is called bounded in a set $X \subseteq Dom(f)$, if $\exists M > 0 \forall x \in Dom(f) |f(x)| < M$.

Theorem 4.12. (Weierstrass theorem)
In a closd interval [a;b] every continuous function is bounded. In this interval the function reaches its maximum and minimum.

Properties of the continuous functions

Theorem 4.13. (Cauchy)
If a function is continuous in a closd interval [a;b], at one endpoint its value is positive, but in other – negative, then at least at one point inside the interval its value is 0.

This theorem has simple geometris interpretation: if the values of the endpoints of the continuous function are with opposite signs, then at least at one point the graph crosses the Ox axis within the interval. At this point the value of the function is 0.

<u>In general, such a result is valid.</u>

Theorem 4.14. (Intermediate value theorem, Cauchy)
Let f be a function defined on $[a;b]$ and let W be a number between $f(a)$ and $f(b)$. If f is continuous on $[a;b]$, then there is at least one number x_0 between a and b such that $f(x_0) = W$.

HOMEWORK 4

7. Please, indicate at what points are the functions discontinuous and according which type. Please substantiate and sketch the graphs.

$$a)\quad y = \begin{cases} \frac{1}{x-1}, x < 0 \\ (x+1)^2, 0 \le x < 2 \\ x^2 - 3, x > 2 \end{cases}$$

$$b)\quad y = \begin{cases} -\frac{1}{x}, x < 0 \\ 5x - x^2, 0 \le x \end{cases}$$

8. Please, indicate if function $g(x) = \frac{|x|}{x-1{,}5}$ is continuous at the given intervals [a;b]; if it is not – please, indicate at what points it is discontinuous and according which type.

a) $[a; b] = [-1; -0{,}5]$, b) $[\mathrm{a}; \mathrm{b}] = [1; 1{,}8]$, c) $[\mathrm{a}; \mathrm{b}] = [2; 4]$.

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p.82 – 92.
Chapter 1. Section 1.6.